



Visualizing genus zero dessins

Mike Musty ICERM 2025 August 12, 2025



Background

- Number theory, computation

Past

- Postdoc (ICERM)
- Mathematician (CRREL)
 - Computer vision, mixed models

Current

- Data wrangler (Faraday)
 - Product implementation
 - Modeling / API infrastructure
 - Client solutions
- Toddler wrangler

Outline



1. Drawings
2. Background
3. Visualizations old and new
4. Examples and motivating questions

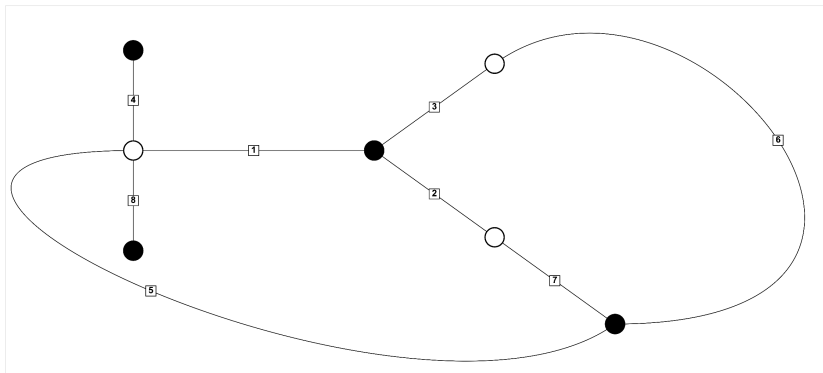
Drawing example



$\sigma_0 = (1,4,5,8)(2,7)(3,6)$, $\sigma_1 = (1,2,3)(5,6,7)$, $\sigma_\infty = (8,7,1)(6,2)(5,4,3)$

Base field: $1 - 3T^2$

Embedding: -1.732051

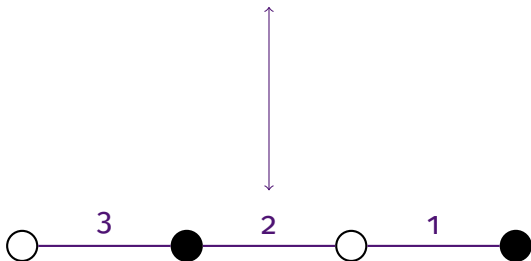


8T38-4.2.2_3.3.1.1_3.3.2

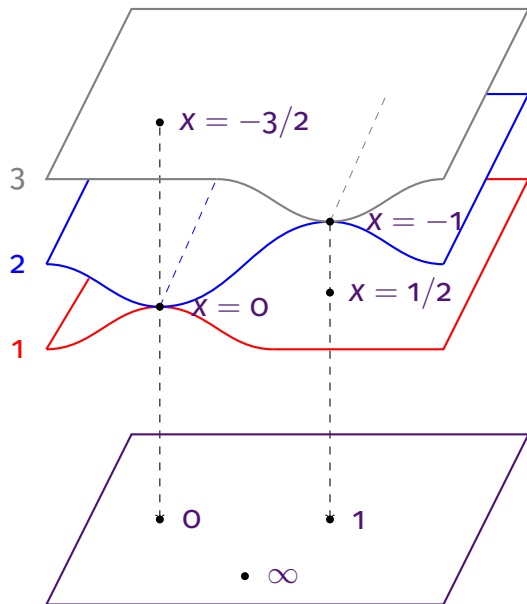
Degree 3 example



$$\left((12)(3), (1)(23), (132) \right)$$



Degree 3 example



$$\begin{aligned} \varphi(x) &= \underbrace{2x^3 + 3x^2}_z \\ &= x^2(2x + 3) \\ &= (2x - 1)(x + 1)^2 + 1 \end{aligned}$$

X

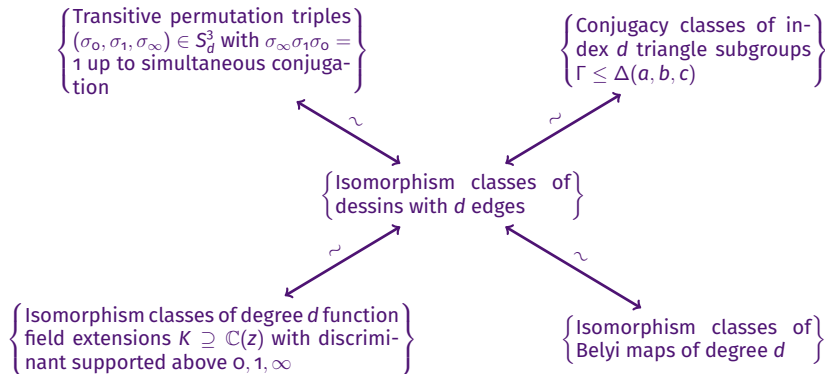
$\mathbb{P}^1$

Belyi maps, dessins, etc.



- This branched covering is an example of a **Belyi map**, a map of Riemann surfaces unbranched outside of $0, 1, \infty$.
- A Belyi map determines a **permutation triple** via monodromy.
- A permutation triple determines a **triangle subgroup** Γ which acts on a spherical, Euclidean, or hyperbolic space $\mathcal{H}$ such that $\Gamma \backslash \mathcal{H} \rightarrow \Delta \backslash \mathcal{H}$ is the corresponding Belyi map.
- The preimage of $[0, 1]$ is a (connected) bipartite graph embedded on a surface called a **dessin d'enfant**.

Equivalent categories

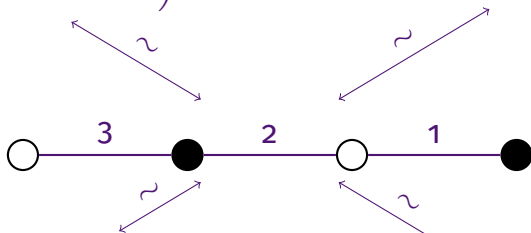


Degree 3 example



$$((12)(3), (1)(23), (132))$$

$$[\Delta(2, 2, 3) : \Gamma] = 3$$



$$\mathbb{C} \left(\underbrace{2x^3 + 3x^2}_z \right) \subseteq \mathbb{C}(x)$$

$$\varphi(x) = \underbrace{2x^3 + 3x^2}_z$$



Theorem (Belyi, 1979)

A smooth projective curve X over $\mathbb{C}$ can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a nonconstant morphism of algebraic curves $\varphi : X \rightarrow \mathbb{P}^1$ unramified outside $\{0, 1, \infty\}$.

In the 1980s Grothendieck described an action of the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the set of (isomorphism classes of) dessins.

Passports



A **passport** consists of the data (G, λ) where

- $G \leq S_d$ is a transitive subgroup
- $\lambda = (\lambda_0, \lambda_1, \lambda_\infty)$ is a triple of partitions of d

A permutation triple $\sigma \in S_d^3$ **belongs** to (G, λ) if

- $\langle \sigma_0, \sigma_1, \sigma_\infty \rangle = G$
- $\sigma_0, \sigma_1, \sigma_\infty$ have the cycle types specified by $\lambda_0, \lambda_1, \lambda_\infty$

Note

- The **size** of a passport is the number of permutation triples belonging to it (up to simultaneous conjugacy).
- A **refined passport** replaces λ with $C = (C_0, C_1, C_\infty)$ a triple of conjugacy classes of G .
- Each (refined) passport is stable under the Galois action and therefore is a union of Galois orbits.

Objective of this work



Draw topologically correct dessins to help visualize the Galois orbits of all (genus zero) passports in the LMFDB.

- <https://www.lmfdb.org/Belyi/>
- <https://michaelmusty.github.io/dessins/>



- <https://www.lmfdb.org/Belyi/7T6/5.1.1/4.2.1/3.2.2/>
- https://michaelmusty.github.io/dessins/passports/7T6-5.1.1_4.2.1_3.2.2/index.html
- All drawings together



- <https://www.lmfdb.org/Belyi/5T4/5/5/3.1.1/>
- https://michaelmusty.github.io/dessins/passports/5T4-5_5_3.1.1/index.html
- All drawings together



- <https://www.lmfdb.org/Belyi/9T34/6.1.1.1/5.2.2/6.1.1.1/>
- https://michaelmusty.github.io/dessins/passports/9T34-6.1.1.1_5.2.2_6.1.1.1/index.html
- All drawings together

Motivating future questions



The ultimate goal is to learn something about the absolute Galois group through its action on dessins. To that end:

- Can we tell when 2 dessins correspond to Galois conjugate Belyi maps?
- Is there topological or combinatorial data that can tell us when a passport splits into multiple Galois orbits?
- Can we tell when a passport is a union of multiple refined passports?

Completeness of the data



$d \backslash g$	0	1	2	3	≥ 4	total
1	1/1	0	0	0	0	1/1
2	1/1	0	0	0	0	1/1
3	2/2	1/1	0	0	0	3/3
4	6/6	2/2	0	0	0	8/8
5	12/12	6/6	2/2	0	0	20/20
6	38/38	29/29	7/7	0	0	74/74
7	89/89	50/50	7/13	2/3	0	148/155
8	243/261	83/217	0/84	0/11	0	326/573
9	410/583	33/427	0/163	0/28	0/6	443/1207
total	802/993	204/732	16/269	2/42	0/6	1024/2042

Thanks!



To:

- Sam Schiavone
<https://math.mit.edu/~sschiavo/>
- Aaron Fenyes <https://ooo.fareycircles.ooo/>
- And to you for listening!